

Team Number: _____ School & Team Name: _____

Student Names: _____

IMPOUND

1. T F 4.a.i Team impounds one Scrambler (with falling mass detached), any alignment devices, additional/spare parts, any papers, and Kit Modification Form (if used). **If Vehicle not impounded, 5000 points added to Final Score.**

CHECK-IN

2. T F 2.c. **Eye Protection:** Participants properly wear Class B eye protection at all times. (If false, **do not run** and give Participation Points)
3. T F 3.j **Kits:** If kits are used, team makes at least two functional modifications during the build process that is documented on the Kit Modification Form. **If F, 50 points added to Run Score.**

Construction Parameters:

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| <u>T</u> | <u>F</u> | 3.a. Components: The Scrambler consist of an egg transport (Vehicle) and an energy propulsion system (launcher and falling mass). |
| <u>T</u> | <u>F</u> | 3.b. Size: The entire Scrambler including the egg, falling mass, and cushion to protect the floor, in the ready-to-run configuration, completely fits within an imaginary rectangular box with a 100.0 cm x 50.0 cm base and a 100.0 cm height. No part of the Scrambler is taped or attached to the event floor. |
| <u>T</u> | <u>F</u> | 3.c. Energy: All energy used to propel the Vehicle comes from a falling mass not exceeding 2.00 kg. Additional sources of kinetic energy are in their lowest energy state in the ready-to-run configuration. Pre-loaded energy storage devices do not provide kinetic energy to propel the vehicle. |
| <u>T</u> | <u>F</u> | 3.c.iii Cushion: Team uses a pad or similar protective cushion to prevent the falling mass from contacting or damaging the floor. The falling mass does not contact the venue floor at any time during the run. |
| <u>T</u> | <u>F</u> | 3.d. Stopping: The stopping mechanism is contained completely within the Vehicle and works automatically. The Vehicle is not remotely controlled or tethered. |
| <u>T</u> | <u>F</u> | 3.e. Backstop and Dowels: Egg rests on two ¼ " - ¾ " round wooden dowels extending between 3.0-4.0 cm from the wooden backstop. Bottom of wooden dowels are 5.0-10.0 cm above the track and within 1.0 cm from the bottom of the backstop. Egg backstop is built of one piece of solid, rigid, flat, and unpadded wood (No Balsa) & has a flat surface of 5.0±0.5 cm wide, 5.0±0.5 cm high, and at least 1.1 cm thick, and mounted rigidly to the front of the vehicle. |
| <u>T</u> | <u>F</u> | 3.h. Electronics: No electric or electronic components are used on the Scrambler. (With the exception of a calculator). |
| <u>T</u> | <u>F</u> | 3.i. Knowledge: Team can answer questions on design, construction, and operation of the device per the Building Policy found on www.soinc.org |

1ST RUN 2ND RUN ← EVENT TIME (8 MINUTES FOR UP TO 2 RUNS)

Construction Parameters:

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| <u>T</u> | <u>F</u> | <u>T</u> | <u>F</u> | ALL CHECK-IN CONSTRUCTION PARAMETERS ARE MET (Re-check construction parameters if necessary). |
| <u>T</u> | <u>F</u> | <u>T</u> | <u>F</u> | 3.f. Actuation: The Scrambler must start by using any part of an unsharpened #2 pencil with an unused eraser provided by the ES, and actuating the release mechanism does not impact additional energy to the Vehicle. |
| <u>T</u> | <u>F</u> | <u>T</u> | <u>F</u> | 3.g. Structural Integrity: All parts of the Vehicle move as a whole; no anchors, tethers, tie downs, or other separate pieces are used. The only parts to contact the floor during the run are those already in contact with the floor in the ready-to-run configuration. All wheels are in contact with the floor at launch. No pieces fall off during the run. No part of the Scrambler can be taped or attached to the event floor. |

4. T F 10. T F ALL CONSTRUCTION PARAMETERS ARE MET. If F, 300 points added to Run Score.

Competition Parameters:

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| <u>T</u> | <u>F</u> | <u>T</u> | <u>F</u> | 4.c.iv No Outside Assistance: Once participants enter the event area to compete, they do not leave or receive outside assistance, materials, or communication until they are finished competing and have left the event area. |
| <u>T</u> | <u>F</u> | <u>T</u> | <u>F</u> | 4.d.iii Securing the egg: The egg will be secured to the dowels on the Backstop with tape. No tape is placed on the front or rear 1.0 cm of the egg. The egg's rounded end is touching the Backstop and visible to the ES when attached. The egg is the foremost point of the Vehicle. |
| <u>T</u> | <u>F</u> | <u>T</u> | <u>F</u> | 4.d.vii Rolling: Team does not roll the Vehicle on the floor of Track on the day of the event or prior day without tournament permission, besides to roll the Vehicle into the Launcher for the sole purpose of loading the Vehicle. |
| <u>T</u> | <u>F</u> | <u>T</u> | <u>F</u> | 4.d.iv Aiming: Only non-electronic sighting/aiming devices are used within the defined Track area. All items not part of the Scrambler are moved and placed behind the Start Point prior to starting a run. |
| <u>T</u> | <u>F</u> | <u>T</u> | <u>F</u> | 4.d.viii Start: In the ready-to-run configuration, the pointed tip of the egg is placed directly above the Start Point within approximately 1 cm. The Scrambler remains in the ready-to-run configuration without being touched until triggered by the #2 pencil. Participants then use a #2 unsharpened pencil to trigger the scrambler and only contact the pencil and not the scrambler during the triggering action. |
| <u>T</u> | <u>F</u> | <u>T</u> | <u>F</u> | 4.d.vi Track: Any substances applied to the device are approved by the Event Supervisor prior to use and do not damage or leave residue on the floor, Track, and/or event area. The Track remains dry at all times. No tape is used on the event floor. |
| <u>T</u> | <u>F</u> | <u>T</u> | <u>F</u> | 4.d.viii.10.d Following Vehicle: Once the run starts, team does not follow their Vehicle and waits until called by the ES to retrieve it. |
| <u>T</u> | <u>F</u> | <u>T</u> | <u>F</u> | 4.d.viii.10.c Terminal Barrier: No parts of the Vehicle (besides maybe the egg) touch the Terminal Barrier. The tip of the egg does not go past the front face of the TB anytime during the run. |
| <u>T</u> | <u>F</u> | <u>T</u> | <u>F</u> | 4.d.viii.10.a Egg: The Egg is not broken during the run. (If false on run 1, 2 nd run will be a failed run) |

5. T F 11. T F ALL COMPETITION PARAMETERS ARE MET. If F, 150 points added to Run Score.

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| 6. SUCCESS | 12. SUCCESS | 8.g. Successful Run: Any run when the Vehicle moves before the end of the Event Time and is NOT a failed run. A failed run is 1) any run where Vehicle starts before ES is ready 2) its distance cannot be measured 3) team pushes the Vehicle down track 4) any run that does not occur in the 8 minutes 5) If egg breaks on first run, second run = failed run. |
| FAILED | FAILED | |

7. _____ 13. _____ 8.e. **Run Time:** Begins when the team actuates the Vehicle and ends when the Vehicle comes to a complete stop.

8. _____ 14. _____ 8.d. **Vehicle Distance:** point-to-point distance, in centimeters to the nearest 0.1 cm, from the Target Point to the Vehicle Measurement Point.

9. _____ 15. _____ 4.viii.5 (Optional) **Inside Can Distance:** Distance between the outer can placed by the ES and the inside can placed by the competitors. (Only fill out if bonus is awarded)

16. T F General Rule: The team is disqualified. (Notify the team and their coach as soon as possible.)